

# EFF-Dock: Fragment-Based SE(3)-Equivariant Flow Matching for Protein–Ligand Docking

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## Abstract

Protein–ligand docking couples pose generation, geometric validity and candidate selection. We present EFF-Dock, a fixed-receptor, supplied-pocket method that learns a flow over rigid-fragment translations and rotations. An equivariant protein–ligand network predicts atom-wise vectors, a Newton–Euler readout projects them onto fragment motion, and separate energy refinement and confidence ranking produce the selected pose. Across five reference-pocket redocking cohorts, refined 100-pose banks contained an RMSD < 2 Å candidate for 87.9–96.5% of complexes. Refinement and rescoring increased selected-pose RMSD–PoseBusters success by 15.7–23.2 percentage points under chirality-filtered selection, yielding success rates of 53.2–80.0%. Ranking within the effective eligible set failed for 12.4–34.4% of complexes despite an available RMSD-successful candidate. These results separate candidate availability, coordinate validity and selection loss, identifying ranking as a major remaining bottleneck in the evaluated pipeline.

A ligand’s three-dimensional pose determines its protein contacts and informs structure–activity analysis. Docking predicts this pose when an experimental complex is unavailable. Here we study fixed-receptor docking with a supplied binding site.

Docking couples global placement and internal conformation on a multimodal landscape. Search must resolve translation, orientation and rotatable bonds while maintaining covalent geometry and protein–ligand complementarity. Receptor preparation, protonation and tautomeric state, cofactors and solvent shape this task. Classical programs combine conformational search with approximate scoring [1, 2]; GNINA adds learned three-dimensional convolutional scoring [3].

Learned geometric models address several receptor and pocket settings. DiffDock samples translation, rotation and torsion [4], Uni-Mol Docking V2 predicts poses within a specified pocket [5], and DiffBindFR includes selected receptor-side-chain flexibility [6]. DynamicBind generates ligand-specific receptor conformations [7], FlowDock uses conditional geometric flow matching [8], and AlphaFold 3 predicts biomolecular complexes by cofolding [9]. We compare supplied-pocket docking pipelines directly and report cofolding references with their distinct input conditions.

Candidate generation, ranking and coordinate validation answer different questions. Full-bank minimum RMSD measures candidate coverage; selected-pose RMSD measures ranking accuracy; and PoseBusters tests stereochemistry, covalent geometry and contacts [10]. Reporting these endpoints separately distinguishes useful sampling from successful deployment. Our confidence network ranks poses within a candidate bank, following the separation of pose-quality and binding-oriented prediction [11].

Rigid-fragment models occupy a middle scale between a rigid ligand and unconstrained atom coordinates. Torsion-tree methods preserve local geometry but move a convention-dependent downstream subgraph at each rotation. Classical incremental docking anchors and grows rigid pieces [12], whereas Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) defines retrosynthetically meaningful fragments [13]. SigmaDock applies diffusion to rigid-fragment poses [14]. EFF-Dock cuts a restricted set of rotatable bonds, moves the resulting components simultaneously, and uses soft geometric terms to maintain their interfaces.

Flow matching learns a vector field that transports a prior toward the target distribution without integrating the generative ordinary differential equation (ODE) during training [15]. Riemannian formulations accommodate translation–rotation spaces [16], including multistage docking approaches such as Matcha [17]. Equivariant tensor products couple scalar, vector and higher-order features under rigid motion [18]. Newton–Euler mechanics has also been used to map interaction energy to flexible-docking motion [19]. In EFF-Dock, the learned network predicts atom-wise vectors and a kinematic readout aggregates them into fragment velocities; a separate interaction energy supplies refinement gradients.

EFF-Dock combines a heterogeneous protein–ligand graph, fragment-space SE(3) flow, geometric refinement and confidence ranking. We evaluate the resulting pipeline across five reference-pocket redocking cohorts, using candidate availability, physical validity and selection loss to locate its remaining bottlenecks.

## 1 Results

### 1.1 Fragment-space docking pipeline

EFF-Dock represents a ligand as rigid components obtained by cutting a restricted set of rotatable bonds. A heterogeneous protein–ligand graph conditions a flow over the fragments’ translations and rotations (Fig. 1). An equivariant network predicts one vector per ligand atom; a least-squares rigid-motion projection converts these vectors to fragment velocities (Fig. 2). A separate network predicts a within-bank pose-ranking score. Generated candidates are refined by bounded descent on a differentiable geometric interaction energy and then ranked again. The energy supplies coordinate gradients independently of the learned vector field; it is neither an affinity predictor nor a free-energy model.

Evaluation uses a fixed holo receptor, a reference-derived pocket, 100 candidates and ten integration steps, with three repeats of the same trained weights. Online Methods and Supplementary Information, Secs. S1.2–S1.3 specify the evaluated coordinate-to-feature conventions and their differences from training.

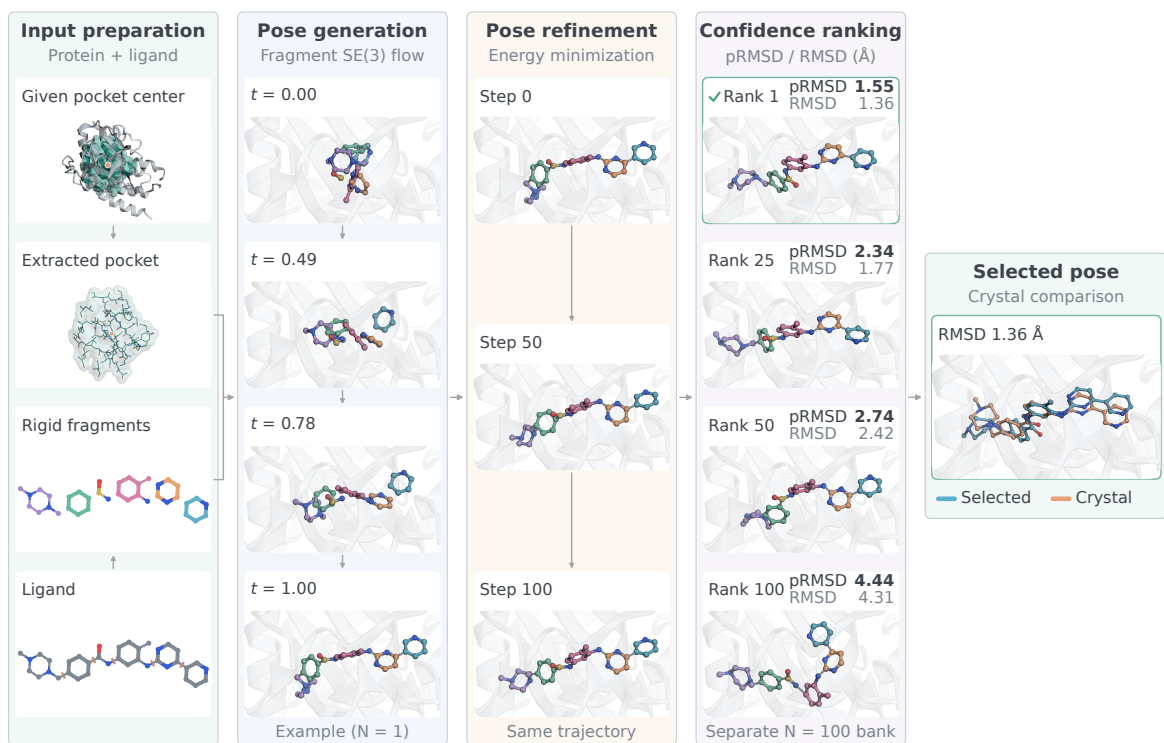
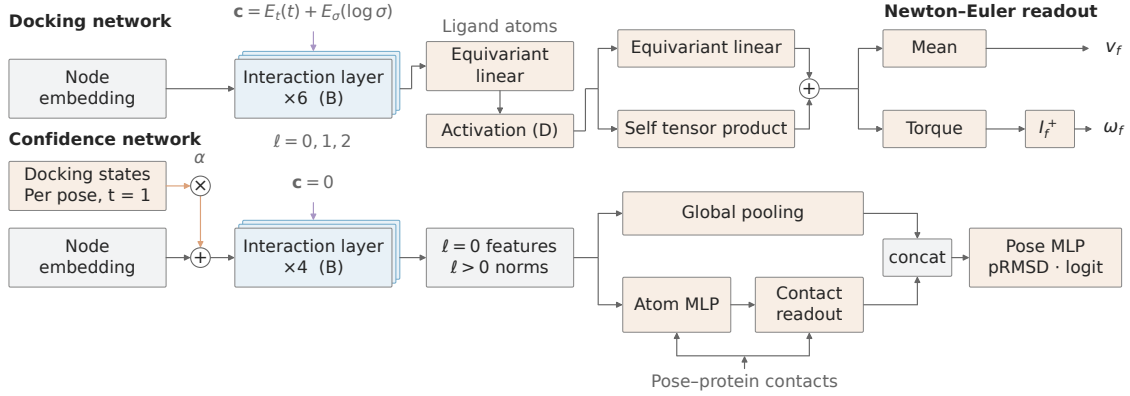
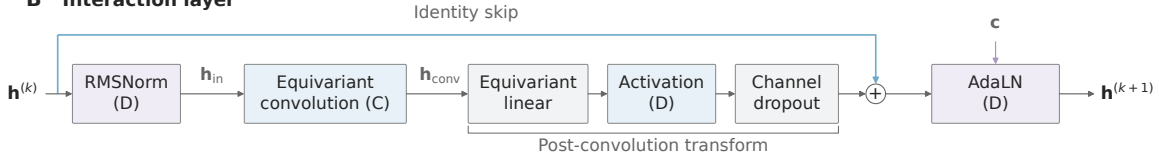


Figure 1: **Overview of the EFF-Dock docking pipeline.** A prepared receptor, ligand chemical state and conformer, and supplied pocket center define the input. Rigid ligand fragments follow the learned SE(3) flow and bounded energy refinement. The illustrated seed-42 trajectory is a prespecified one-pose example from a supplied conformer. Confidence ranking uses a separate 100-pose benchmark bank prepared from SMILES. Ranks use predicted RMSD (pRMSD); displayed RMSD and crystal overlays are retrospective evaluations. The checkmark identifies the selected candidate and crosses unselected examples.

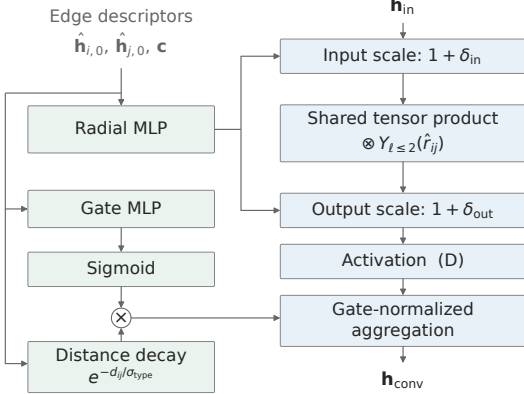
### A Model architecture



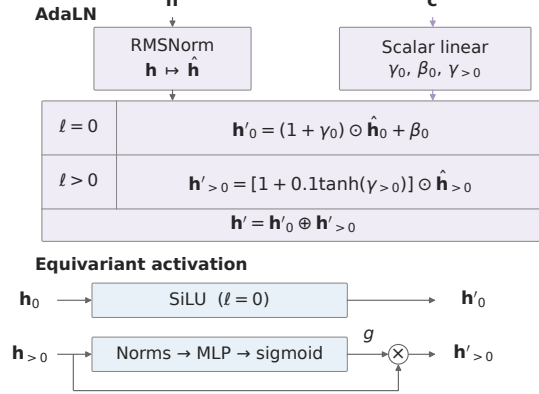
### B Interaction layer



### C Equivariant convolution



### D Normalization and activation



**Figure 2: Docking and confidence architectures.** (A) The docking network applies six conditioned equivariant interaction layers and converts ligand-atom features to fragment translation and angular velocities with the Newton–Euler readout. For each candidate, terminal docking features are recomputed at  $t = 1$  with its source prior scale. The confidence network combines these with a new embedding, applies four interaction layers with zero conditioning vector  $\mathbf{c} = 0$ , invariantizes the resulting features, and predicts pose-level RMSD and success scores from global and contact-aware summaries. (B) Each interaction layer contains normalization, equivariant convolution, post-convolution transformation, a residual connection, and adaptive layer normalization. (C) The equivariant convolution uses radial and gating networks, shared tensor products through  $\ell \leq 2$ , distance decay, and gate-normalized aggregation with invariant nonscalar rescaling. (D) Scalar and nonscalar channels use type-appropriate adaptive normalization and equivariant activation. Detailed tensor definitions are provided in Supplementary Information, Sec. S3.

## 1.2 Supplied-pocket comparison

Fig. 3 compares selected-pose RMSD success and RMSD–PoseBusters success on Astex Diverse and PoseBusters v2. The local comparison includes SigmaDock [14], DiffDock-Pocket [20], RLDiff RL++ [21], DiffBindFR [6], and force-optimized SurfDock [22]. On Astex, EFF-Dock achieved 80.0% same-pose RMSD–PoseBusters success. SigmaDock, SurfDock and RLDiff RL++ had higher point estimates on both endpoints. On PoseBusters v2, EFF-Dock reached 79.1% RMSD–PoseBusters success. Relative to SigmaDock, the paired difference in RMSD–PoseBusters success was +2.7 percentage points (95% bootstrap interval,  $-2.3$  to  $+7.5$ ), compared with  $-9.0$  points ( $-16.1$  to  $-2.0$ ) on Astex. The PoseBusters interval includes zero, leaving relative performance unresolved; intervals are exploratory and unadjusted for multiple comparisons (Supplementary Fig. S14). Most selected RMSD-successful EFF-Dock poses also passed PoseBusters.

Learned methods retain their native preparation, sampling and ranking procedures; GOLD and Vina are literature values [10]. Protocol and candidate-budget differences are specified in Supplementary Information, Sec. S7.4.

Supplementary Table S14 and Fig. S20 extend the comparison to PhiBench-derived, FoldBench and the official 802-case OpenBind subset using benchmark-specific endpoints. They include pocket-conditioned baselines, cofolding references and GNINA holo redocking, with task, cohort, preparation and evaluability differences specified in Sec. S7.3.

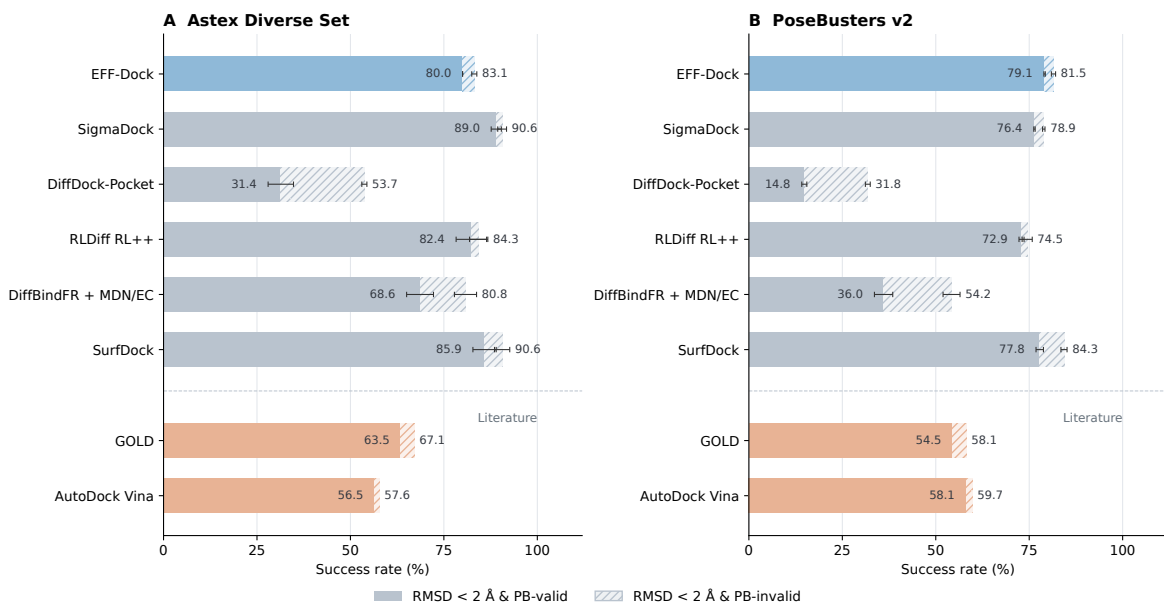


Figure 3: **External docking performance on supplied-pocket benchmarks.** (A) Astex Diverse Set (85 complexes) and (B) PoseBusters v2 (308 complexes). Solid bars show RMSD  $< 2$  Å and PoseBusters success; hatched segments complete RMSD-only success. Local results are means with sample SD across three repeats. EFF-Dock uses 100 poses, 10 sampling steps, refinement, and chirality-filtered confidence selection. GOLD and AutoDock Vina are literature values; the remaining methods use their native sampling and ranking protocols. DiffDock-Pocket uses a holo-aligned predicted receptor; DiffBindFR samples pocket side-chain torsions on a fixed backbone and uses cognate-ligand coordinates in its upstream scoring graph; its selected ligand poses are evaluated against the common frozen holo receptor. Budgets, receptor preparation, selectors and evaluation versions differ, so these are protocol-conditioned comparisons.

## 1.3 Refinement and chirality-aware selection

Fig. 4 compares refinement and chirality-aware selection. Refinement and rescoring increased mean RMSD–PoseBusters success by 15.7–23.2 percentage points under the primary selector and 21.7–26.5

points without chirality filtering. Without chirality filtering, mean RMSD-only selected-pose success increased more modestly. These gains combine coordinate refinement with confidence reselection.

Chirality filtering added smaller mean improvements of 0.3–2.4 percentage points after refinement (Supplementary Fig. S9). Refinement also increased input-tetrahedral compatibility across all banks, addressing stereochemistry at the cut interfaces between internally rigid fragments.

The complexity analysis in Supplementary Fig. S1 localizes the remaining error. Selected top-1 and RMSD–PoseBusters success generally deteriorated for larger ligands, more rotatable bonds, and more fragments, most clearly in PoseBusters, PhiBench-derived, and FoldBench. Oracle success was substantially less sensitive than selected top-1, so the top-1–oracle gap tended to widen in harder strata. Molecular complexity affects ranking and selected-pose validity as well as near-native pose generation. The smallest Astex and high-complexity bins are too sparse for a clear trend.

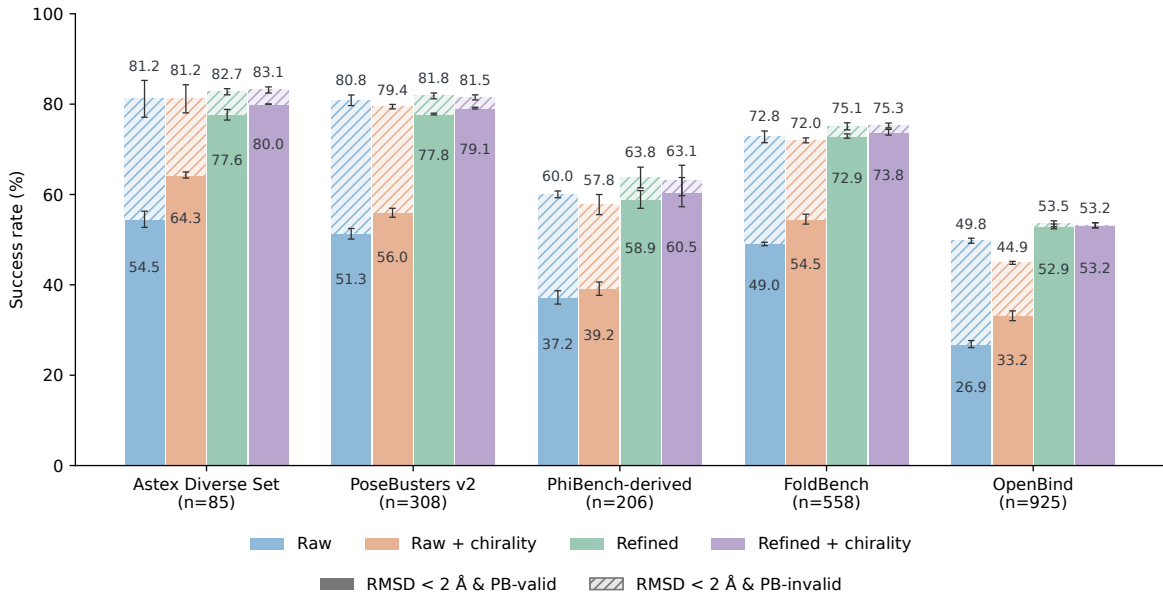


Figure 4: **Effects of refinement and chirality-aware selection.** EFF-Dock results are shown for Astex, PoseBusters v2, PhiBench-derived, FoldBench, and OpenBind using the frozen released weights and unguided 100-pose, 10-step sampling. Raw and refined banks are ranked with or without the chirality filter. Solid bars show RMSD and PoseBusters success; hatched segments complete RMSD-only success. Bars and error bars are the mean and sample SD across three repeats on fixed cohorts.

Supplementary Fig. S16 resolves the aggregate validity gain into individual PoseBusters checks. Refinement reduced selected-pose failures associated with bond geometry, internal clashes, receptor clashes, and internal energy, while stereochemical and ring-planarity failures were less uniformly affected. The matched-candidate panels separate same-index fail-to-pass and pass-to-fail transitions from changes caused by selecting a different pose after refinement. Matched-index transitions characterize the subset with saved official evaluations at both stages (Supplementary Fig. S16).

## 1.4 Confidence ranking and selection headroom

Fig. 5 evaluates the ranker within each generated bank. For refined banks, mean within-complex Spearman correlation between predicted and observed RMSD ranged from 0.51 to 0.70; near-native classification AUROC ranged from 0.88 to 0.94 and average precision from 0.61 to 0.82. The median selection regret and conditional top-1 success in Fig. 5C,D quantify how an available RMSD-successful candidate can fail to be selected. Reliability curves and performance as a function of near-native candidate density are reported in Supplementary Fig. S12.

Confidence ranking enriches near-native poses at early ranks relative to generation order (Supplementary Fig. S2). In refined unfiltered banks, expanding from top-1 to top-5 increased RMSD coverage by 7.5–20.3 percentage points, although substantial gaps to the full-bank oracle remained (Supplemen-

tary Fig. S10). OpenBind illustrates this selection bottleneck: only 6.3% of its refined candidates were near-native, yet 87.9% of complexes contained at least one such pose.

The generation-prefix experiment in Supplementary Fig. S3 reaches the same conclusion from a deployable selection perspective. Oracle coverage increases or stays constant as candidates are added; selected top-1 can decline when the ranker promotes an incorrect new pose. This non-monotonicity persists in some chirality-filtered prefixes. Larger candidate banks improve coverage, but the gain reaches top-1 only when the ranker remains stable as new poses are added.

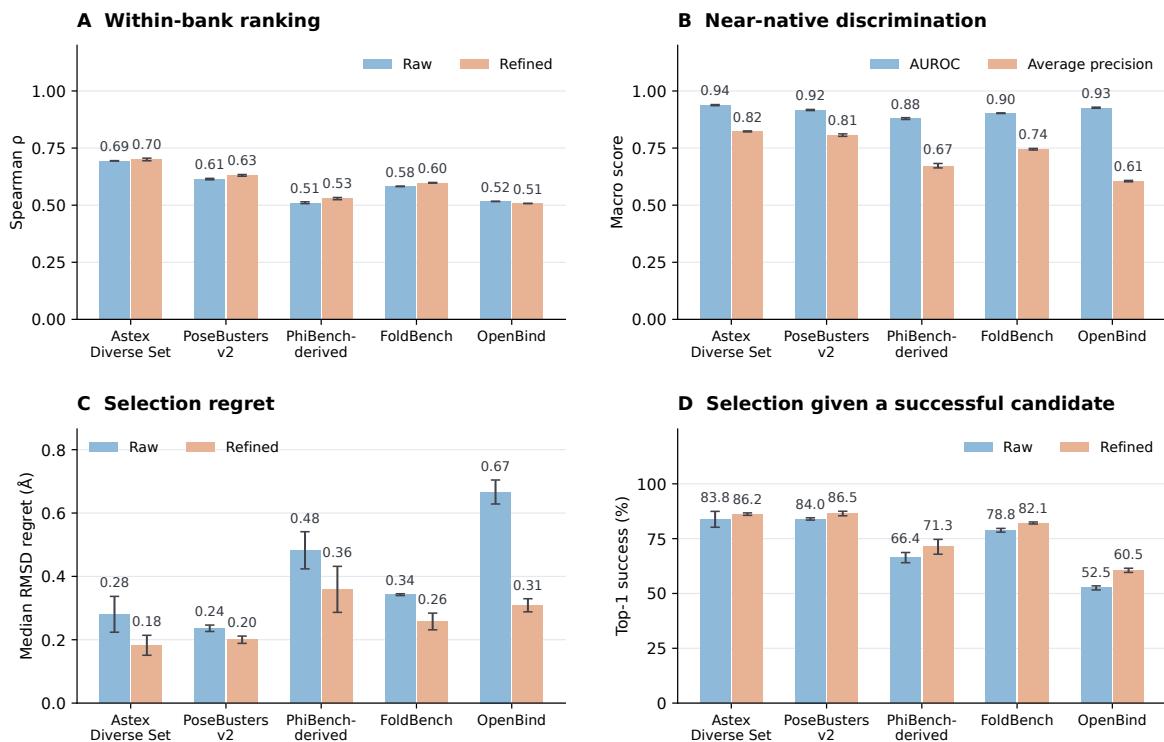


Figure 5: **Confidence ranking and remaining selection loss.** (A) Within-complex Spearman correlation between predicted and observed RMSD for raw and refined banks. (B) Refined-bank AUROC and average precision for strict RMSD < 2 Å labels; single-class banks are excluded. (C) Median selected-minus-full-oracle RMSD regret. (D) Selected top-1 success conditional on the full bank containing a near-native candidate. Panels C and D use the primary chirality-filtered selector, so their loss can include eligibility exclusions. Bars show three-repeat means and sample SD.

## 1.5 Eligibility-aware selection bottlenecks

Supplementary Fig. S18 partitions every primary outcome into absence of an RMSD-successful candidate in the refined bank, exclusion of all near-native candidates by the effective chirality mask, ranking failure within the eligible set, PoseBusters failure of an RMSD-successful selection, or RMSD–PoseBusters success. Chirality exclusion accounted for at most 0.3% of complexes. Ranking failure was larger, ranging from 12.4% on PoseBusters to 34.4% on OpenBind. RMSD-based ranking failure exceeds chirality exclusion in the refined banks. Its joint-validity headroom remains unmeasured because PoseBusters labels cover selected candidates. The regret curves in Supplementary Fig. S18B measure loss relative to the best RMSD in the same effective candidate set available to the deployed selector.

Training-set relatedness is examined in Supplementary Figs. S6–S8. Sequence and ligand overlap differed substantially among cohorts, but neither exact ligand matches nor sequence-identity strata showed a consistent association with docking success. The relatedness strata contextualize these cohort-specific outcomes.

## 1.6 Sensitivity to generation inputs

We varied the generation crop, supplied-center error and translation-prior scale on all Astex and PoseBusters complexes while holding the networks, downstream spatial context and primary selector fixed (Fig. 6). With a 10 Å crop and prior scale 2 Å, RMSD–PoseBusters success was 80.0% on Astex and 79.1% on PoseBusters without center error. Center error with standard deviation 2 Å per axis reduced success to 71.0% and 64.4%, respectively. Refined-bank RMSD-only oracle coverage at 2 Å error was 86.3% and 82.8%.

Larger generation crops partly buffered center error, whereas prior-scale effects varied with the cohort and perturbation level. Fig. 6 reports the complete grids under the fixed primary protocol.

Refinement and confidence scoring retained the original unperturbed center and a 10 Å crop in all conditions. The crop and center-jitter grids perturb sampling inputs with fixed downstream spatial context. Prior-scale comparisons include the scale’s influence on both sampling and the recomputed docking features used for ranking. Guidance and candidate-allocation controls at equal learned pose-step counts are reported separately (Supplementary Fig. S19).

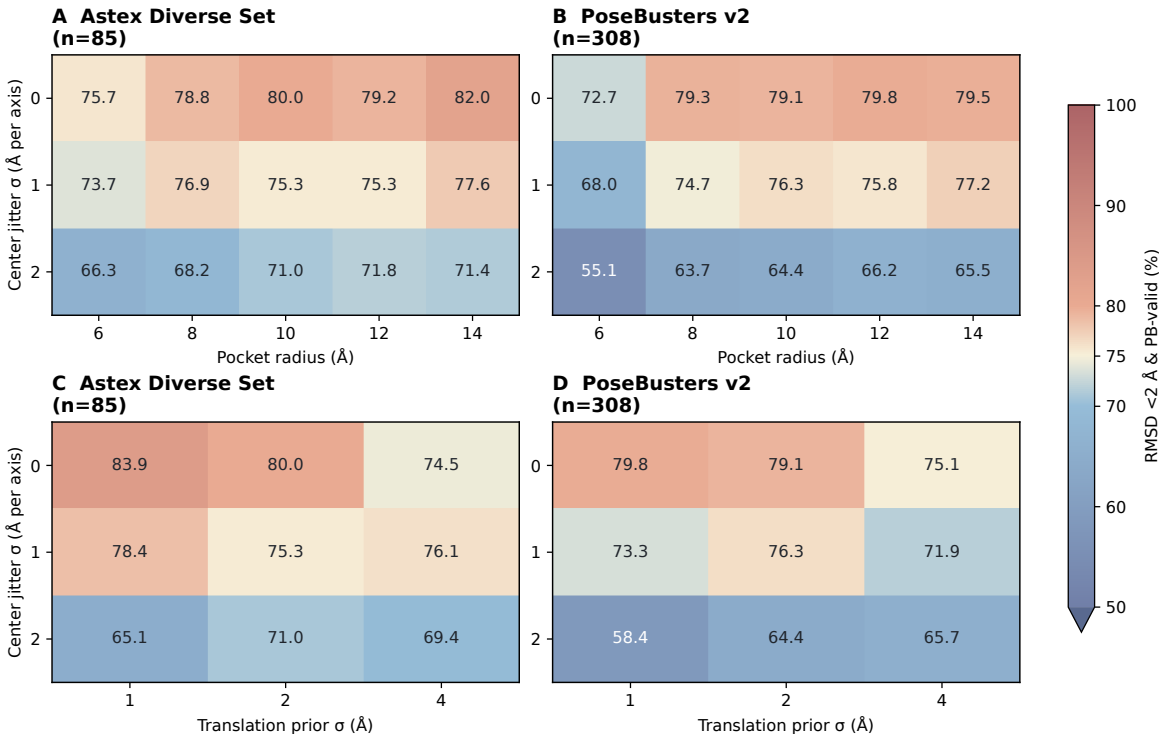


Figure 6: **Sensitivity to supplied-pocket radius, center error and translation-prior scale.** (A,B) Joint selected-pose RMSD < 2 Å and PoseBusters success across generation crop radii at prior scale 2 Å, for Astex Diverse Set (85 complexes) and PoseBusters v2 (308 complexes). (C,D) Prior-scale dependence at a 10 Å generation crop on the same cohorts. Rows vary Gaussian center error, with standard deviation reported per Cartesian axis. Cells show three-inference-seed means. All conditions use unguided 100-pose, ten-step generation, energy refinement and chirality-filtered confidence ranking. Refinement and ranking retain the original center and 10 Å crop; terminal docking features used for confidence retain the generation prior scale. The grid isolates the crop and prior axes rather than crossing every pair. Downstream spatial context is fixed; prior-scale effects include sampling and terminal feature extraction.

## 2 Discussion

Refinement and ranking improve different stages of docking. Refinement with rescoring raised RMSD–PoseBusters success more than RMSD-only success, emphasizing geometric validity in the aggregate gain. Matched-candidate transitions support coordinate repair within the evaluated subset, while primary selected-pose improvements also include reselection. The remaining RMSD oracle-top-1 gap measures selection loss; its joint-validity headroom requires PoseBusters labels beyond selected candidates.

Candidate density clarifies this gap. OpenBind combined sparse near-native candidates with high full-bank coverage, and complex ligands were more susceptible to ranking loss. Adding candidates expanded coverage but could reduce top-1 accuracy when an incorrect pose was promoted. Ranking failure within the effective eligible set exceeded chirality exclusion (Supplementary Fig. S18), motivating rankers that remain stable as candidate banks grow.

The experiments assess the complete fixed-receptor, supplied-pocket pipeline. Component contributions remain unresolved without controlled ablations, and within-bank confidence diagnostics establish empirical ranking behavior rather than cross-generator calibration. Three fixed-weight repeats measure sampling variation. Evaluation-specific coordinate-to-feature handling differs from training; finite tests support joint-world covariance conditional on the prepared body convention, with residual unobservable-twist dependence documented in Online Methods and Supplementary Information, Secs. S1.2–S1.3.

The scope is known-pocket holo redocking with one ligand chemical state and starting conformer. Geometry training uses crystal-derived pockets and fragment templates; rigid-fragment motion preserves intrafragment errors, and systematic starting-conformer sensitivity remains unmeasured. Center perturbations vary generation inputs with downstream spatial context fixed. Receptor flexibility, pocket discovery, solvent rearrangement, chemical-state enumeration and covalent docking are separate tasks; unsupported cofactors, metal clusters and boundary-case macrocycles remain outside the demonstrated chemistry. Fragmentation uses a fixed atom ordering and soft cut-interface constraints. The interaction energy represents geometric preferences and the confidence score within-bank pose quality; pose accuracy and validity are distinct from binding or affinity.

Generalization depends on the evaluated cohorts and training exposure. Astex and PoseBusters were used repeatedly during development, PhiBench-derived and FoldBench adapt their source tasks, and OpenBind concentrates one protease family, including two noncovalent approximations of covalent systems. Training relatedness is substantial: all OpenBind cases occupy the highest sequence-identity bin, and many FoldBench cases share a training accession (Supplementary Figs. S6–S8 and Table S16). Accession sharing differs from complex identity, and the ancestral exposure inventory is incomplete. Only 22 external complexes satisfy the simultaneous sequence, ligand-similarity and exact-match restrictions; broader unrelated-target evaluation remains necessary.

EFF-Dock separates rigid-fragment generation, geometric refinement and candidate ranking within one docking pipeline. Its finite banks provide broad near-native coverage under the evaluated preparation, while selection loss identifies a concrete target for improving deployment accuracy.

## 3 Online Methods

### 3.1 Coordinate convention and evaluated weights

The coordinate reconstruction in Eq. (2) uses an active local-to-world rotation,  $\mathbf{x} = \mathbf{R}\mathbf{y} + \mathbf{T}$ . Equivariance is defined conditional on a fixed prepared fragment-local template and its body-coordinate convention. Under a joint rigid transformation of the receptor and current ligand pose, the prepared local coordinates remain fixed while world coordinates transform as  $\mathbf{x}' = \mathbf{Q}\mathbf{x} + \mathbf{a}$  and fragment rotations as  $\mathbf{R}' = \mathbf{Q}\mathbf{R}$ . Supplied conformers are centered fragment-wise without canonicalizing their local axes. Accordingly, a learned fragment-local vector  $\mathbf{w}$  is injected into world-frame features as

$$\mathbf{h} = \mathbf{R}\mathbf{w}, \quad \mathbf{h}' = (\mathbf{Q}\mathbf{R})\mathbf{w} = \mathbf{Q}\mathbf{h} \tag{1}$$

under a joint world rotation  $\mathbf{Q} \in \text{SO}(3)$ . Source-local edge coordinates of the form  $\mathbf{R}^\top(\mathbf{x}_j - \mathbf{x}_i)$  perform the inverse world-to-local map.

All measurements use fixed docking and confidence weights; no retraining, probability recalibration, or result-dependent thresholding was performed. Coordinate-to-feature conventions, their differences from training, and finite numerical verification are documented in the [Supplementary Information, Sec. S1.2](#).

### 3.2 Task definition and fragment pose representation

EFF-Dock takes a prepared fixed receptor  $\mathcal{P}$ , a heavy-atom ligand graph  $\mathcal{G}_L$  in one specified chemical state, one supplied or generated ligand conformer  $\mathcal{C}_L$ , and an explicit pocket center  $\mathbf{b} \in \mathbb{R}^3$ . We denote their complete conditioning context by  $\mathcal{C}_{\text{cond}} = (\mathcal{P}, \mathcal{G}_L, \mathcal{C}_L, \mathbf{b})$ . It returns a finite set of ligand poses and a separate score used only to rank poses within that set. Deployment inputs exclude target ligand coordinates, target-derived frames, RMSD, validity outcomes and benchmark identity. The reference-pocket experiments derive the supplied center from the reference structure before removing the ligand.

The ligand contains  $N$  heavy atoms partitioned into  $K$  rigid fragments, with  $i \in \{1, \dots, N\}$  and  $k \in \{1, \dots, K\}$  indexing atoms and fragments, respectively. The map  $g(i)$  assigns atom  $i$  to a fragment, and  $\mathbf{y}_i \in \mathbb{R}^3$  is its local coordinate relative to that fragment’s geometric centroid. The generative state and atom reconstruction are

$$\mathcal{M}_L = \prod_{k=1}^K (\mathbb{R}^3 \times \text{SO}(3)), \quad \mathbf{z} = \{(\mathbf{T}_k, \mathbf{R}_k)\}_{k=1}^K, \quad \mathbf{x}_i(\mathbf{z}) = \mathbf{R}_{g(i)}\mathbf{y}_i + \mathbf{T}_{g(i)}. \quad (2)$$

Here  $\mathcal{M}_L$  is the ligand pose space,  $\mathbb{R}^3$  is the translation space, and  $\text{SO}(3)$  is the group of proper three-dimensional rotations. The group  $\text{SE}(3)$  combines proper rotations and translations. Coordinates and distances use Angstrom, angles use radians unless degrees are shown, and flow time is dimensionless. Superscript  $\top$  denotes transpose and  $\mathbf{I}_3$  is the three-dimensional identity matrix; further notation and parameter conventions are collected in the [Supplementary Information, Sec. S1.1](#). For fragment  $k$ ,  $\mathbf{T}_k \in \mathbb{R}^3$  is its geometric-centroid state origin and  $\mathbf{R}_k \in \text{SO}(3)$  is its rotation;  $\mathbf{z}$  collects all fragment states, and  $\mathbf{x}_i(\mathbf{z}) \in \mathbb{R}^3$  is the reconstructed coordinate of ligand atom  $i$  in the pocket-centered frame. This representation preserves intrafragment geometry while allowing fragments to translate and rotate relative to one another.

Ligands are standardized with RDKit [23]: stereochemistry is assigned before hydrogen removal, the largest connected component is retained, and formal charge and the supplied protonation state are preserved. For simplified molecular-input line-entry system (SMILES) input, the inference path constructs one seeded Experimental-Torsion Knowledge Distance Geometry version 3 (ETKDGv3) conformer [24], followed by at most 200 Merck Molecular Force Field (MMFF) iterations [25]; supplied structure-data file (SDF) or Tripos MOL2 coordinates are retained. The receptor is hydrogen-stripped and fixed, and all receptor and ligand coordinates are translated by  $-\mathbf{b}$  so that the supplied pocket center is the origin. For geometry training, each deposited crystal ligand pose is fragmented in its bound conformation, and the resulting crystal coordinates define the fixed fragment-local coordinates used as model input and reconstruction templates. Prospective inputs instead use supplied or independently generated conformers; rigid-fragment motion cannot correct internal differences within a fragment. Exact input normalization and pocket provenance are given in the [Supplementary Information, Sec. S1](#).

### 3.3 Fragmentation and pocket-conditioned graph

Fragmentation cuts eligible heavy-atom single bonds outside rings while excluding planar carbonyl C–N/O and sulfonyl S–N linkages. Connected components define provisional rigid fragments, singleton components are merged into adjacent fragments, and the retained cut topology supplies fragment adjacency and local cross-fragment distance constraints. This indexed greedy rule is deterministic for a fixed atom ordering; exact eligibility, tie-breaking, and failure handling are specified in the [Supplementary Information, Sec. S2.1](#).

The receptor is cropped residue-wise around the supplied center. Geometry training samples a radius uniformly from 6 to 12 Å and applies isotropic center jitter with standard deviation 2 Å; held-out validation uses an 8 Å crop, and the external reference-pocket protocol uses a fixed 10 Å crop. The graph contains ligand atoms, ligand fragments, protein atoms, and residue virtual nodes. Typed static relations encode covalent structure, fragment membership and topology, protein hierarchy, and

residue–fragment context, while protein-atom–ligand-atom contacts within 5 Å are rebuilt at every forward pass. Node chemistry, all ten directed relation types, crop fallbacks, and edge descriptors are listed in the [Supplementary Information, Sec. S3.1](#).

### 3.4 Equivariant fragment flow and Newton–Euler readout

After pocket centering, fragment translations are drawn from an isotropic Gaussian and multi-atom fragment rotations from the uniform distribution on  $\text{SO}(3)$  [26]. Let  $t \in [0, 1]$  denote dimensionless flow time, with subscripts 0 and 1 denoting the sampled prior and bound target states, respectively, and let  $\sigma > 0$  be the translation-prior standard deviation in Angstrom. For a paired prior and target state, translations follow a linear path and rotations follow the shortest geodesic under a left-action convention:

$$\begin{aligned} \mathbf{T}_{k,0} &\sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3), & \mathbf{R}_{k,0} &\sim \text{Uniform}(\text{SO}(3)), \\ \mathbf{T}_{k,t} &= (1-t)\mathbf{T}_{k,0} + t\mathbf{T}_{k,1}, & \mathbf{R}_{k,t} &= \text{Exp}(t\boldsymbol{\omega}_k^*)\mathbf{R}_{k,0}, \\ \mathbf{v}_k^* &= \mathbf{T}_{k,1} - \mathbf{T}_{k,0}, & \boldsymbol{\omega}_k^* &= \text{Log}(\mathbf{R}_{k,1}\mathbf{R}_{k,0}^{-1}). \end{aligned} \quad (3)$$

Here  $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3)$  is the three-dimensional Gaussian prior and  $\text{Uniform}(\text{SO}(3))$  is the normalized Haar distribution over rotations. The maps  $\text{Exp} : \mathbb{R}^3 \rightarrow \text{SO}(3)$  and  $\text{Log} : \text{SO}(3) \rightarrow \mathbb{R}^3$  are the rotation exponential and principal logarithm under the axis–angle identification. The starred vectors  $\mathbf{v}_k^*$  and  $\boldsymbol{\omega}_k^*$  are the target translational and angular tangents, measured in Angstrom and radians per unit flow time, respectively. Singleton rotations are fixed to identity and contribute no supervised angular mode. At time  $t$ ,  $\mathbf{z}_t = \{(\mathbf{T}_{k,t}, \mathbf{R}_{k,t})\}_{k=1}^K$  denotes the interpolated ligand state. Conditional flow matching trains the vector field  $u_\theta(\mathbf{z}_t, t, \mathcal{C}_{\text{cond}}, \sigma)$ , with learned parameters  $\theta$ , to return the fragment-wise predictions  $\{(\hat{\mathbf{v}}_k, \hat{\boldsymbol{\omega}}_k)\}_{k=1}^K$  and match the starred targets without integrating the generative ODE during training [15]. The prior-scale mixture, time distribution, rotation augmentation, and numerical path conventions are recorded in the [Supplementary Information, Sec. S2](#).

**Equivariant trunk and Newton–Euler readout.** All node types share six equivariant interaction layers with 384 scalar channels, 32 channels of each vector parity and 16 channels of each rank-two parity. Time and prior scale enter through a 128-dimensional conditioning embedding. Each layer combines normalized sender features, radial edge features and real spherical harmonics through order two using tensor products, gated residual aggregation and adaptive normalization. Static relations and rebuilt protein–ligand contacts use relation-specific edge features. Complete equations and dimensions are given in the [Supplementary Information, Sec. S3](#).

The output head predicts one polar atom-vector proposal  $\mathbf{f}_i$  for each ligand atom. Its unit-weight Newton–Euler aggregation is a least-squares rigid-body velocity fit restricted to the retained rotational modes; these vectors are not physical forces or energy gradients. For fragment  $k$ , with atom set  $\mathcal{A}_k$ , center  $\mathbf{T}_k$ , and lever arm  $\mathbf{r}_i = \mathbf{x}_i - \mathbf{T}_k$ , the fragment field is obtained by

$$\begin{aligned} \hat{\mathbf{v}}_k &= \frac{1}{|\mathcal{A}_k|} \sum_{i \in \mathcal{A}_k} \mathbf{f}_i, & \boldsymbol{\mu}_k &= \sum_{i \in \mathcal{A}_k} \mathbf{r}_i \times \mathbf{f}_i, \\ \mathbf{J}_k &= \sum_{i \in \mathcal{A}_k} (\|\mathbf{r}_i\|_2^2 \mathbf{I}_3 - \mathbf{r}_i \mathbf{r}_i^\top), & \hat{\boldsymbol{\omega}}_k &= \mathbf{J}_k^+ \boldsymbol{\mu}_k. \end{aligned} \quad (4)$$

Here  $\hat{\mathbf{v}}_k$  is the predicted translational tangent,  $\boldsymbol{\mu}_k$  is the moment of the atom vectors,  $\mathbf{J}_k$  is the unit-mass geometric inertia about the fragment centroid, and  $\hat{\boldsymbol{\omega}}_k$  is the predicted angular tangent. The truncated pseudoinverse  $\mathbf{J}_k^+$  removes geometric-inertia eigenmodes at or below 1% of the largest eigenvalue, and the same observable subspace is applied to the angular training target. This prevents singleton and nearly collinear fragments from receiving arbitrary angular velocities; the exact eigenspace construction is given in the [Supplementary Information, Sec. S3.2](#).

### 3.5 Objective and sampling

The primary losses are mean-squared errors (MSEs) on fragment translation and observable angular velocity. The geometry objective combines fragment translation and observable angular-velocity errors

with two auxiliary terms:

$$\mathcal{L}_{\text{geom}} = \mathcal{L}_v + \lambda_\omega \mathcal{L}_\omega + \lambda_{\text{atom}} \mathcal{L}_{\text{atom}} + \lambda_{\text{DG}} \mathcal{L}_{\text{DG}}. \quad (5)$$

Here  $\mathcal{L}_v$  is fragment translation MSE,  $\mathcal{L}_\omega$  is angular MSE after projecting the target onto retained rotational modes, and  $\mathcal{L}_{\text{atom}}$  compares reconstructed atom velocities. The distance-geometry term  $\mathcal{L}_{\text{DG}}$  compares interfragment atom-pair distances after a one-step endpoint extrapolation, with a  $t^2$ -weighted valid-complex reduction. Unobservable terms contribute zero. The final continuation uses weights  $(\lambda_\omega, \lambda_{\text{atom}}, \lambda_{\text{DG}}) = (8, 0.3, 3)$ ; definitions, normalizations and distributed reductions are given in the [Supplementary Information, Sec. S3.3](#).

The final continuation samples flow time from an 80% base distribution, a 10% uniform component on  $[0, 0.3]$ , and a 10% point mass at  $t = 0$ . The base distribution mixes uniform sampling with a late-biased logit-normal distribution. Its parameters and the prior-scale mixture are specified in the [Supplementary Information, Sec. S2](#).

At inference, sampled fragment priors are integrated from  $t = 0$  to 1 using 10 late-biased Euler intervals in the primary protocol. For interval width  $\Delta t$ , translations receive additive increments and rotations are advanced by left multiplication with  $\text{Exp}(\Delta t \tilde{\omega}_k)$ , where the tilde denotes the guided field when guidance is enabled. Atom coordinates and dynamic protein-ligand contacts are rebuilt before every field evaluation. The sampling schedule and candidate selection are specified in the [Supplementary Information, Sec. S2.2](#).

### 3.6 Confidence-based pose selection

Pose generation and top-1 selection are separated. For each candidate, the fixed docking network is evaluated at  $t = 1$ , using recovered fragment frames and the source translation-prior scale, to recompute terminal ligand representations. A second equivariant network combines these with a fresh heterogeneous-graph embedding and applies four additional interaction layers. Invariantized node features and explicit protein-contact summaries are pooled by ligand-contact attention and type-wise mean-maximum aggregation. Atom and pose heads are trained with displacement and RMSD regression, 2 Å success classification, pairwise ranking, and a success-focused listwise objective. For a candidate set  $\{\mathbf{z}_S^{(m)}\}_{m=1}^M$ , the deployed selector is

$$\mathcal{E} = \begin{cases} \{m : q_m = 1\}, & \text{if any } q_m = 1, \\ \{1, \dots, M\}, & \text{otherwise,} \end{cases} \quad m^* = \arg \min_{m \in \mathcal{E}} c_\phi(\mathbf{z}_S^{(m)}, \mathcal{P}, \mathcal{G}_L, \mathcal{C}_L, \sigma). \quad (6)$$

Here  $q_m \in \{0, 1\}$  indicates whether candidate  $m$  matches all assigned input tetrahedral stereocenters, and  $\mathcal{E}$  is the effective eligible set. Ties follow saved candidate order. The unfiltered argmin is reported separately as a diagnostic condition. The confidence network’s nonnegative predicted-RMSD score is  $c_\phi$ , with learned parameters  $\phi$ ;  $m^*$  is the selected candidate index. The prior scale enters this score through the recomputed docking representations. The score ranks candidates within a bank; empirical reliability is evaluated separately from the deployed selector. Energy restraints and confidence features use conformers prepared by the same procedure and seed as pose generation. Refinement preserves each saved candidate’s fragment-internal coordinates. For rescoring, candidate coordinates are mapped to fragment frames by proper Kabsch fitting for templates of rank at least two and an input-indexed, scene-defined convention for lower-rank fragments. This resolves the recovery convention without making the frozen network invariant to unobservable fragment twist; finite numerical checks and their limits are reported in the [Supplementary Information, Sec. S1.3](#). The complete readout and confidence objective are given in the [Supplementary Information, Sec. S4](#).

### 3.7 Differentiable guidance and post-sampling refinement

For ligand coordinates  $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$ , we define a differentiable guidance energy

$$\begin{aligned} E_G &= E_{\text{phys}} + E_{\text{int}}, \\ E_{\text{phys}} &= E_{\text{bond}} + E_{\text{angle}} + E_{\text{proper}} + E_{\text{chiral}} + E_{\text{planar}} \\ &\quad + E_{\text{LJ}} + E_{\text{steric}} + E_{\text{obs}}, \\ E_{\text{int}} &= E_{\text{hyd}} + E_{\text{HB}} + E_{\text{chg}} + E_\pi + E_{\text{cat-}\pi} + E_{\text{XB}} + E_{\text{metal}}. \end{aligned} \quad (7)$$

The cut-interface terms restrain only bond, angle, proper-torsion, chiral, and planar geometry that spans independently moving fragments. The nonbonded terms include a soft-core Universal Force Field (UFF)-style Lennard-Jones profile on admitted interfragment ligand-ligand and protein-ligand pairs, a compact receptor-overlap barrier and repulsion-only obstacles for unresolved receptor chemistry [27]. The seven interaction terms describe hydrophobic contacts, directional hydrogen bonds, screened formal-charge groups, pi stacking, cation-pi contacts, halogen bonds, and typed metal coordination [28, 29, 30, 31, 32]. These terms define a geometric refinement objective in nominal energy units.

Compact radial and angular gates use a clipped quintic smoothstep with zero first and second derivatives at its endpoints. The envelope joins constant regions with continuous energy and first and second derivatives. Energy forms, occupancy aggregation, typing conventions and numerical constants are specified in the [Supplementary Information, Sec. S5](#).

The coordinate gradient  $\nabla_{\mathbf{x}_i}$  gives the guidance force  $\mathbf{F}_i = -\nabla_{\mathbf{x}_i} E_G$ , which is distinct from the learned atom-vector proposal  $\mathbf{f}_i$ . For atom mass  $m_i$ , the fragment mass is  $m_k^{\text{frag}} = \sum_{i \in \mathcal{A}_k} m_i$ , the center of mass is  $\mathbf{c}_k = (m_k^{\text{frag}})^{-1} \sum_{i \in \mathcal{A}_k} m_i \mathbf{x}_i$ , and the resultant is  $\mathbf{F}_k = \sum_{i \in \mathcal{A}_k} \mathbf{F}_i$ . The vector  $\boldsymbol{\tau}_k = \sum_{i \in \mathcal{A}_k} (\mathbf{x}_i - \mathbf{c}_k) \times \mathbf{F}_i$  is the center-of-mass torque, and  $\mathbf{I}_k$  is the corresponding mass-weighted inertia tensor. The translational and angular guidance directions are

$$\mathbf{d}_k^\omega = \mathbf{I}_k^+ \boldsymbol{\tau}_k, \quad \mathbf{d}_k^T = \frac{\mathbf{F}_k}{m_k^{\text{frag}}} + \mathbf{d}_k^\omega \times (\mathbf{T}_k - \mathbf{c}_k). \quad (8)$$

Here  $\mathbf{d}_k^T$  and  $\mathbf{d}_k^\omega$  are guidance directions,  $\mathbf{I}_k^+$  is the mass-inertia truncated pseudoinverse, and  $\mathbf{T}_k - \mathbf{c}_k$  corrects for the offset between the fragment’s geometric-centroid state origin and its center of mass. For guidance experiments, let  $(\Delta \mathbf{v}_k, \Delta \boldsymbol{\omega}_k)$  denote the mass-inertia directions after RMS matching to the learned atom velocity, late-time interval ramping, and the translation, rotation, and atom-displacement caps. The guided field is

$$\tilde{\mathbf{v}}_k = \hat{\mathbf{v}}_k + \Delta \mathbf{v}_k, \quad \tilde{\boldsymbol{\omega}}_k = \hat{\boldsymbol{\omega}}_k + \Delta \boldsymbol{\omega}_k, \quad (9)$$

where  $\eta$  controls the pre-cap guidance strength; the exact interval average, normalization, zero-direction rule, and caps are given in the [Supplementary Information, Sec. S5.9](#). This inference-time modification is conceptually related to guidance of flow-matching vector fields, although the normalization and fragment projection used here are method specific [33]. Post-sampling refinement updates only the fragment state of each candidate. At refinement iteration  $r$ ,

$$\mathbf{T}_k^{(r+1)} = \mathbf{T}_k^{(r)} + \alpha_r \bar{\mathbf{d}}_k^T, \quad \mathbf{R}_k^{(r+1)} = \text{Exp}(\alpha_r \bar{\mathbf{d}}_k^\omega) \mathbf{R}_k^{(r)}, \quad (10)$$

Here  $(\bar{\mathbf{d}}^T, \bar{\mathbf{d}}^\omega) = \text{Cap}(\mathbf{d}^T, \mathbf{d}^\omega)$  are capped increments and  $\alpha_r$  is the accepted line-search step size. The cap first bounds each fragment translation and rotation separately, then jointly rescales both until the reconstructed atom displacement meets its bound. Writing  $E_r = E_G(\mathbf{x}^{(r)})$ , the proposed translation, rotation, and induced atom displacement are bounded and  $\alpha_r$  is reduced by monotone backtracking until a finite non-increasing guidance energy is obtained. The refinement is bounded and stops on convergence, line-search failure, or a sustained energy plateau; the numerical safeguards and termination criteria are given in the [Supplementary Information, Sec. S5.10](#). Raw and refined candidates are compared with unchanged network parameters.

The guidance energy uses a typed 18 Å receptor shell, separate from the 10 Å neural crop. Full typing rules, energy equations, mass-inertia projection, refinement safeguards, and permitted information are provided in the [Supplementary Information, Sec. S5](#).

### 3.8 Data curation and split

Training data were derived from PLINDER 2024-06/v2 [34]. Curation applies ligand validity, covalency, interaction, experimental-quality, resolution, contact and size criteria, followed by deterministic representative selection within ligand and pocket-community groups. The filters and cohort counts are detailed in the [Supplementary Information, Sec. S6](#).

After model-specific input filtering, docking uses 47,277 training and 1,076 validation samples, while confidence uses 43,092 training and 1,035 validation samples. Their differing input requirements and overlap are detailed in the [Supplementary Information, Sec. S6](#).

External exclusion required both an exact ligand match and membership in a benchmark-associated pocket community, removing 904 samples. Recurring ligands and homologous proteins are assessed separately through training-relatedness analysis.

### 3.9 Evaluation scope

Row-level splits can share repeated ligands or homologous proteins, so training-set relatedness is measured explicitly rather than inferred from split labels [34, 35, 36, 37]. For each external cohort, we report the receptor state, pocket source, ligand preparation, candidate budget, and selector. Astex and PoseBusters were used during development and are treated as descriptive benchmarks.

The geometry model uses the receptor crop, ligand graph, starting conformer, pocket center, and current sampled coordinates. The confidence model ranks the generated candidates from the same task inputs. The interaction energy may additionally read a larger receptor shell, but it has no access to the crystal pose, RMSD, or validity labels. Raw coordinates are saved before refinement.

### 3.10 Model training and selection

The docking model was trained for 200,000 base updates, 100,000 geometry-fine-tuning updates and 50,000 early-time-replay updates, with global batch sizes of 256, 256 and 64. The final stage used AdamW with peak learning rate  $2 \times 10^{-5}$ , weight decay 0.01, gradient clipping at 1.0 and exponential moving average (EMA) decay 0.999 [38]; its terminal EMA was evaluated. The confidence model was trained on raw and refined poses and selected at update 70,000 by predicted-RMSD-ranked top-1 success on 1,035 internal-validation complexes. Training stages, initialization, optimization schedules and selection criteria are specified in the [Supplementary Information, Sec. S6.4](#). External benchmark results did not select the evaluated checkpoints.

### 3.11 Benchmarks and comparison protocol

The evaluation uses five complementary cohorts. The Astex Diverse Set contains 85 curated complexes [39], and PoseBusters v2 contains 308 complexes with detailed coordinate-validity checks [10]. The PhiBench-derived cohort comprises 206 locally selected complexes from the public PhysDock archive; identity with the published 206-case cohort has not been established [40]. FoldBench-Pocket contains 558 released protein-ligand interfaces, including a 66-interface post-June 2024 slice [41]. OpenBind contains 925 EV-A71/CVA16 2A-protease complexes and is an auxiliary single-family cohort, including two flagged noncovalent approximations of covalent systems [42].

All five cohorts are evaluated by holo-receptor, reference-pocket redocking. The pocket center is computed from receptor residue nodes within 8 Å of the reference ligand, and the reference ligand is then removed from the model input.

The primary setting uses 100 candidates, 10 late-biased Euler steps,  $\sigma = 2$  Å, a 10 Å receptor crop, and no inference-time guidance. Each cohort uses the same trained weights with inference seeds 42, 100042 and 200042. Astex and PoseBusters fix the ETKDG seed at zero for generation; PhiBench-derived, FoldBench and OpenBind use the per-complex sampling seed for conformer generation, so their repeat variability also includes starting-conformer variation. Candidates are scored before and after bounded refinement. The primary selector removes chirality-mismatched candidates and then chooses the lowest predicted symmetry-aware RMSD; if none passes the chirality screen, it uses the unfiltered ranking. Failed complexes remain in the cohort denominator. Supplementary Figs. S4 and S5 report runtime and generation-process memory for the candidate-acquisition runs, including their original refinement and confidence scoring but excluding subsequent refinement and rescoring passes. Their scope is initial candidate acquisition, as defined in the figure captions. Sensitivity to generation inputs is analyzed in Fig. 6; supplementary guidance controls are described below.

### 3.12 Generation-input sensitivity and guidance controls

Sensitivity experiments use all 85 Astex and 308 PoseBusters complexes with the same three inference seeds and chirality-filtered selector. At translation-prior scale 2 Å, generation crop radii 6, 8, 10, 12, 14 Å are crossed with Gaussian center jitter of 0, 1, 2 Å per Cartesian axis. Separately, prior scales 1, 2, 4 Å are crossed with the same jitter levels at a 10 Å crop. Shared cells are evaluated once, and the primary baseline is reused from the primary evaluation. The jitter random stream is independent of the pose prior, and matching seeds couple prior draws within a candidate budget. Refinement and confidence scoring always use the original center and a 10 Å crop; the docking-feature re-evaluation used for confidence scoring retains each candidate’s source prior scale.

Four budget/guidance conditions, including the shared primary baseline, cross 100 poses/ten steps or 40 poses/25 steps with unguided sampling or normalized energy guidance of strength  $\eta = 2$ . Both budgets use 1,000 learned pose-steps, but guidance and candidate processing add work, so equal pose-step counts do not imply equal runtime. Forty- and 100-pose banks are generated separately under prespecified control settings.

### 3.13 Endpoints and statistical analysis

The primary pose endpoint is confidence-selected, symmetry-aware heavy-atom RMSD below 2 Å after complete atom mapping. Secondary endpoints include top- $q$  and oracle- $q$  coverage and complete-case RMSD summaries. Physical validity uses PoseBusters 0.6.5 in **redock** mode, with a reference-molecule representation adapter for the three temporal cohorts as detailed in the [Supplementary Information, Sec. S7](#); pass-all is the conjunction of all 27 non-RMSD checks on the same selected pose [10]. RMSD–PoseBusters success requires both RMSD < 2 Å and PoseBusters pass-all.

Binary endpoints use the full cohort denominator; continuous summaries use successfully evaluated poses and report their coverage. Error bars are sample standard deviations across the three repeats unless stated otherwise. Supplementary Figs. S9 and S14 use paired 95% bootstrap intervals. Further evaluation details are given in the [Supplementary Information, Sec. S7](#).

### 3.14 Use of automated tools

OpenAI Codex and Anthropic Claude were used to assist with code review, analysis-script development, manuscript drafting, consistency checks and language editing. The authors reviewed the generated text against the source code and primary literature and retain full responsibility for the manuscript. These tools are not authors.

## Data and Code Availability

Source code, model weights, training-membership lists, numerical source data and analysis scripts are available at <https://github.com/eightmm/EFF-Dock>. Benchmark structures remain subject to their original distribution and redistribution terms.

## Author Contributions

J.S.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and writing—original draft. J.L.: conceptualization, methodology, resources, supervision, project administration, and writing—review and editing. Both authors are responsible for the final scientific content.

## Competing Interests

J.L.’s affiliation with Arontier Co. is disclosed in the author information.

## Supplementary Information

Supplementary methods and analyses are provided in the accompanying Supplementary Information.

## References

- [1] Oleg Trott and Arthur J. Olson. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2):455–461, 2010.

- [2] Jerome Eberhardt, Diogo Santos-Martins, Andreas F. Tillack, and Stefano Forli. AutoDock Vina 1.2.0: New docking methods, expanded force field, and python bindings. *Journal of Chemical Information and Modeling*, 61(8):3891–3898, 2021.
- [3] Andrew T. McNutt, Yanjing Li, Rocco Meli, Rishal Aggarwal, and David Ryan Koes. GNINA 1.3: The next increment in molecular docking with deep learning. *Journal of Cheminformatics*, 17:28, 2025.
- [4] Gabriele Corso, Hannes Stärk, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. DiffDock: Diffusion steps, twists, and turns for molecular docking. In *International Conference on Learning Representations*, 2023.
- [5] Eric Alcaide, Zhifeng Gao, Guolin Ke, Yaqi Li, Linfeng Zhang, Hang Zheng, and Gengmo Zhou. Uni-Mol Docking V2: Towards realistic and accurate binding pose prediction. *arXiv preprint arXiv:2405.11769*, 2024.
- [6] Jintao Zhu, Zhonghui Gu, Jianfeng Pei, and Luhua Lai. DiffBindFR: An SE(3) equivariant network for flexible protein–ligand docking. *Chemical Science*, 15(21):7926–7942, 2024.
- [7] Wei Lu, Jixian Zhang, Weifeng Huang, Ziqiao Zhang, Xiangyu Jia, Zhenyu Wang, Leilei Shi, Chengtao Li, Peter G. Wolynes, Shuangjia Zheng, et al. DynamicBind: Predicting ligand-specific protein–ligand complex structure with a deep equivariant generative model. *Nature Communications*, 15:1071, 2024.
- [8] Alex Morehead and Jianlin Cheng. FlowDock: Geometric flow matching for generative protein–ligand docking and affinity prediction. *Bioinformatics*, 41(Supplement 1):i198–i206, 2025.
- [9] Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, John M. Jumper, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 2024.
- [10] Martin Buttenschoen, Garrett M. Morris, and Charlotte M. Deane. PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences. *Chemical Science*, 15:3130–3139, 2024.
- [11] Jaemin Sim and Juyong Lee. BA-Pred and RMSD-Pred: Integrated graph neural network models for accurate protein–ligand binding affinity and binding pose prediction. *Journal of Chemical Information and Modeling*, 66(7):3480–3495, 2026.
- [12] Matthias Rarey, Bernd Kramer, Thomas Lengauer, and Gerhard Klebe. A fast flexible docking method using an incremental construction algorithm. *Journal of Molecular Biology*, 261(3):470–489, 1996.
- [13] Jörg Degen, Christof Wegscheid-Gerlach, Andrea Zaliani, and Matthias Rarey. On the art of compiling and using ‘Drug-Like’ chemical fragment spaces. *ChemMedChem*, 3(10):1503–1507, 2008.
- [14] Alvaro Prat, Leo Zhang, Charlotte M. Deane, Yee Whye Teh, and Garrett M. Morris. SigmaDock: Untwisting molecular docking with fragment-based SE(3) diffusion. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [15] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- [16] Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *The Twelfth International Conference on Learning Representations*, 2024.
- [17] Daria Frolova, Talgat Daulbaev, Egor Sevriugov, Sergei A. Nikolenko, Dmitry N. Ivankov, Ivan Oseledets, and Marina A. Pak. Matcha: Multi-stage riemannian flow matching for accurate and physically valid molecular docking. *arXiv preprint arXiv:2510.14586*, 2026. Version 3.
- [18] Yi-Lun Liao and Tess Smidt. Equiformer: Equivariant graph attention transformer for 3d atomistic graphs. In *International Conference on Learning Representations*, 2023.

- [19] Yufei Huang, Odin Zhang, Lirong Wu, Cheng Tan, Haitao Lin, Zhangyang Gao, Siyuan Li, and Stan Z. Li. Re-Dock: Towards flexible and realistic molecular docking with diffusion bridge. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 20474–20489. PMLR, 2024.
- [20] Michael Plainer, Marcella Toth, Simon Dobers, Hannes Stärk, Gabriele Corso, Céline Marquet, and Regina Barzilay. DiffDock-Pocket: Diffusion for pocket-level docking with sidechain flexibility. In *Machine Learning in Structural Biology Workshop, NeurIPS*, 2023.
- [21] J. Henry Broster, Bojana Popovic, Diana Kondinskaia, Charlotte M. Deane, and Fergus Imrie. Teaching diffusion models physics: Reinforcement learning for physically valid diffusion-based docking. *Chemical Science*, 2026.
- [22] Duanhua Cao, Mingan Chen, Runze Zhang, Zhaokun Wang, Manlin Huang, Jie Yu, Xinyu Jiang, Zhehuan Fan, Wei Zhang, Hao Zhou, et al. SurfDock is a surface-informed diffusion generative model for reliable and accurate protein–ligand complex prediction. *Nature Methods*, 22:310–322, 2025.
- [23] Greg Landrum et al. RDKit: Open-source cheminformatics software. <https://www.rdkit.org>, 2024.
- [24] Shuzhe Wang, Jagna Witek, Gregory A. Landrum, and Sereina Riniker. Improving conformer generation for small rings and macrocycles based on distance geometry and experimental torsional-angle preferences. *Journal of Chemical Information and Modeling*, 60(4):2044–2058, 2020.
- [25] Thomas A. Halgren. Merck molecular force field. i. basis, form, scope, parameterization, and performance of MMFF94. *Journal of Computational Chemistry*, 17(5–6):490–519, 1996.
- [26] Ken Shoemake. Uniform random rotations. In David Kirk, editor, *Graphics Gems III*, pages 124–132. Academic Press, 1992.
- [27] Anthony K. Rappé, Carla J. Casewit, K. S. Colwell, William A. Goddard, III, and W. M. Skiff. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *Journal of the American Chemical Society*, 114(25):10024–10035, 1992.
- [28] Sebastian Salentin, Sven Schreiber, V. Joachim Haupt, Melissa F. Adasme, and Michael Schroeder. PLIP: Fully automated protein–ligand interaction profiler. *Nucleic Acids Research*, 43(W1):W443–W447, 2015.
- [29] Georgia B. McGaughey, Marc Gagné, and Anthony K. Rappé.  $\pi$ -stacking interactions: Alive and well in proteins. *Journal of Biological Chemistry*, 273(25):15458–15463, 1998.
- [30] Justin P. Gallivan and Dennis A. Dougherty. Cation– $\pi$  interactions in structural biology. *Proceedings of the National Academy of Sciences of the United States of America*, 96(17):9459–9464, 1999.
- [31] Gautam R. Desiraju, P. Shing Ho, Lars Kloo, Anthony C. Legon, Roberto Marquardt, Pierangelo Metrangolo, Peter Politzer, Giuseppe Resnati, and Kari Rissanen. Definition of the halogen bond (IUPAC recommendations 2013). *Pure and Applied Chemistry*, 85(8):1711–1713, 2013.
- [32] Ivan Dokmanić, Mile Šikić, and Sanja Tomić. Metals in proteins: Correlation between the metal ion type, coordination number and the amino-acid residues involved in the coordination. *Acta Crystallographica Section D: Biological Crystallography*, 64(3):257–263, 2008.
- [33] Ruiqi Feng, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. On the guidance of flow matching. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 16993–17029. PMLR, 2025.
- [34] Janani Durairaj, Yusuf Adeshina, Zhonglin Cao, Xuejin Zhang, Vladas Oleinikovas, et al. PLINDER: The protein–ligand interactions dataset and evaluation resource. *bioRxiv*, 2024.

- [35] Gabriele Corso, Arthur Deng, Nicholas Polizzi, Regina Barzilay, and Tommi Jaakkola. Deep confident steps to new pockets: Strategies for docking generalization. In *International Conference on Learning Representations*, 2024.
- [36] Alex Morehead, Nabin Giri, Jian Liu, Pawan Neupane, and Jianlin Cheng. Assessing the potential of deep learning for protein–ligand docking. *Nature Machine Intelligence*, 8:32–41, 2026.
- [37] Peter Škrinjar, Jérôme Eberhardt, Gabriel Studer, Gerardo Tauriello, Torsten Schwede, and Janani Durairaj. Evaluating generalization in protein–ligand cofolding methods. *Nature Structural & Molecular Biology*, 33:782–794, 2026.
- [38] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- [39] Michael J. Hartshorn, Marcel L. Verdonk, Gianni Chessari, Suzanne C. Brewerton, Wijnand T. M. Mooij, Paul N. Mortenson, and Christopher W. Murray. Diverse, high-quality test set for the validation of protein–ligand docking performance. *Journal of Medicinal Chemistry*, 50(4):726–741, 2007.
- [40] Kexin Zhang, Yuanyuan Ma, Jiale Yu, Huiting Luo, Jinyu Lin, Yifan Qin, Xiangcheng Li, Qian Jiang, Fang Bai, Jiayi Dou, et al. PhysDock: A physics-guided all-atom diffusion model for protein–ligand complex prediction. *bioRxiv*, 2025.
- [41] Sheng Xu, Qiantai Feng, Lifeng Qiao, Hao Wu, Tao Shen, Yu Cheng, Shuangjia Zheng, and Siqi Sun. Benchmarking all-atom biomolecular structure prediction with FoldBench. *Nature Communications*, 17:442, 2026.
- [42] OpenBind Consortium. OpenBind structure–affinity data release: Enterovirus A71 (EV-A71) / coxsackievirus A16 (CVA16) 2a protease, 2026.